

Bilinear Groups and BGN

Lecture 9

SEC 595 — Encrypted Computing

- 1 Bilinear Groups
- 2 BGN: Keys, Encryption, Decryption
- 3 Additive Homomorphism
- 4 The One Multiplication
- 5 Security

Lecture 8: ElGamal multiplies, Paillier adds — each *one* operation, unlimited.

BGN (Boneh–Goh–Nissim, 2005)

A **somewhat homomorphic** scheme:

- any number of **additions**;
- exactly **one** multiplication.

The second operation comes from one new object: a **bilinear pairing**.

Definition

Let $\mathbb{G}$, $\mathbb{G}_1$ be multiplicative cyclic groups of order n , and g a generator of $\mathbb{G}$. A map $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_1$ is **bilinear** if

$$e(u^a, v^b) = e(u, v)^{ab} \quad \text{for all } u, v \in \mathbb{G}, a, b \in \mathbb{Z},$$

and $e(g, g)$ generates $\mathbb{G}_1$ (non-degeneracy).

$\mathbb{G}$ is a **bilinear group** if such $\mathbb{G}_1$ and e exist.

Both arguments come from the same group (*symmetric*), so a ciphertext can be paired with a ciphertext. Efficient constructions exist on elliptic curves; we use only the two properties.

What the pairing buys — and costs

$$e(g^a, g^b) = e(g, g)^{ab}$$

Two hidden exponents go in; their **product** comes out — once, because the result lies in $\mathbb{G}_1$ and there is no map out of $\mathbb{G}_1$.

DDH is easy in $\mathbb{G}$

Given g, g^a, g^b, z , test $e(g^a, g^b) \stackrel{?}{=} e(g, z)$. Equal exactly when $z = g^{ab}$.

So BGN cannot rely on DDH; it relies on a *composite* group order.

Example — checking bilinearity

$\mathbb{G} = \langle g \rangle$ cyclic.

- a. From $e(u^a, v^b) = e(u, v)^{ab}$, show $e(u_1 u_2, v) = e(u_1, v) e(u_2, v)$.
- b. Write $e(g^a, g^b) e(g^c, g^d)$ as a power of $e(g, g)$.

The composite-order group

A generator $\mathcal{G}(\tau)$ outputs $(q_1, q_2, \mathbb{G}, \mathbb{G}_1, e)$ with

$$|\mathbb{G}| = |\mathbb{G}_1| = n = q_1 q_2,$$

q_1, q_2 distinct primes of similar size.

The subgroup of order q_1

For a generator u of $\mathbb{G}$, $h = u^{q_2}$ has order exactly q_1 :
 $\text{ord}(u^k) = n / \gcd(n, k) = n / q_2 = q_1$.

Writing h as a power of g : $h = g^{\alpha q_2}$ with $\gcd(\alpha, q_1) = 1$. Only the algebra uses α .

KeyGen(τ)

- 1 Run $\mathcal{G}(\tau)$; $n = q_1 q_2$.
- 2 Random generators $g, u \xleftarrow{R} \mathbb{G}$; set $h = u^{q_2}$.
- 3 $\text{Pk} = (n, \mathbb{G}, \mathbb{G}_1, e, g, h)$, $\text{Sk} = q_1$.

g , order n	carries the <i>message</i>
h , order q_1	carries the <i>mask</i>
q_1	annihilates the mask
q_2	bounds the message space

u and α are not published. Factoring n reveals Sk .

Messages: $m \in \{0, 1, \dots, T\}$ with $T < q_2$; usually $T = 1$.

$\text{Enc}(\text{Pk}, m)$

Pick $r \xleftarrow{R} \{0, \dots, n-1\}$ and output

$$C = \underbrace{g^m}_{\text{message}} \underbrace{h^r}_{\text{mask}} \in \mathbb{G}.$$

One group element. The mask lies in the order- q_1 subgroup; the message does not.

Dec(Sk, C)

$$C^{q_1} = (g^m h^r)^{q_1} = (g^{q_1})^m, \quad m = \log_{g^{q_1}} C^{q_1}.$$

The mask dies: $(h^r)^{q_1} = (u^{q_1 q_2})^r = 1$ for every r . Exact — no noise.

The message is still in an exponent, so a **discrete log** remains. It is easy only because m is tiny: with $T = 1$, check whether C^{q_1} is 1 or g^{q_1} .

Proposition

$\text{ord}(g^{q_1}) = n/q_1 = q_2$. So $(g^{q_1})^m$ depends only on $m \bmod q_2$, and decryption is well defined iff $T < q_2$.

Message space	Decryption
T small (e.g. $T = 1$)	short search: practical
T near q_2	full discrete log: infeasible

A cost condition, not a security condition. BGN encrypts bits or very small integers.

Example — a toy instance

$q_1 = 5$, $q_2 = 7$, $n = 35$. $\mathbb{G}$ is the subgroup of order 35 in $\mathbb{Z}_{71}^*$. Take $g = 2$, $u = 27$.

- a. Compute $h = u^{q_2} \bmod 71$ and check $\text{ord}(h) = 5$.
- b. Find α with $h = g^{\alpha q_2}$. Write Pk and Sk.
- c. Encrypt $m = 0$ and $m = 1$, each with $r = 6$ and $r = 13$.

Example — decrypting the toy ciphertexts

Same parameters: $g = 2$, $h = 25$, $q_1 = 5$, modulo 71.

- a. List the powers of $g^{q_1} = 32$ and confirm $\text{ord}(32) = 7$.
- b. Decrypt the ciphertexts 25, 5, 50, 10.

Multiplying ciphertexts adds messages

$$C_1 C_2 = g^{m_1} h^{r_1} g^{m_2} h^{r_2} = g^{m_1+m_2} h^{r_1+r_2}$$

A fresh-looking ciphertext of $m_1 + m_2$, decrypted by the same q_1 .

No budget: the mask is cancelled exactly, so additions chain without limit.

The plaintext must still fit: Dec returns $m_1 + m_2 \bmod q_2$.

Example — addition and its limit

Toy instance: $g = 2$, $h = 25$, $q_1 = 5$, modulo 71.

- a. $C_1 = \text{Enc}(1; r = 6) = 50$, $C_2 = \text{Enc}(1; r = 3) = 10$. Compute $C_1 C_2$ and decrypt.
- b. With $T = 1$, how should the result be read?
- c. What does the product of seven encryptions of 1 decrypt to?

The target group

Two public elements of $\mathbb{G}_1$:

$$g_1 = e(g, g) \text{ (order } n), \quad h_1 = e(g, h) = g_1^{\alpha q_2} \text{ (order } q_1).$$

$\mathbb{G}_1$ has the same message/mask structure as $\mathbb{G}$, with (g_1, h_1) in place of (g, h) .

Definition

Given $C_1, C_2 \in \mathbb{G}$, pick fresh $r \in \mathbb{Z}_n$ and output

$$C = e(C_1, C_2) h_1^r \in \mathbb{G}_1.$$

With $C_i = g^{m_i + \alpha q_2 r_i}$, bilinearity gives

$$C = g_1^{m_1 m_2} h_1^{\tilde{r}}$$

— every other term of the exponent carries αq_2 , so it lands in $\langle h_1 \rangle$.

Uses only public data; $\tilde{r}$ is never computed.

$$C^{q_1} = (g_1^{q_1})^{m_1 m_2}, \quad m_1 m_2 = \log_{g_1^{q_1}} C^{q_1}.$$

Same private key q_1 ; the mask $h_1^{\tilde{r}}$ dies as before.

Only the base changes, from g^{q_1} to $g_1^{q_1}$ (order q_2). The decryptor must know which group C is in.

Why exactly one multiplication

Inputs live in $\mathbb{G}$; the product lives in $\mathbb{G}_1$. There is no map out of $\mathbb{G}_1$, so it cannot be paired again.

A typing limit

BGN's ceiling is not a noise bound: larger parameters do not raise it.

Product ciphertexts can still be multiplied in $\mathbb{G}_1$, which adds their messages. Supported: **sums of products of pairs** — polynomials of degree ≤ 2 , any number of terms, result $< q_2$.

Example — expanding the exponent

- a. Expand $e(g^{m_1} h^{r_1}, g^{m_2} h^{r_2}) h_1^r$ using $h = g^{\alpha q_2}$, $g_1 = e(g, g)$, $h_1 = e(g, h)$.
- b. Write it as $g_1^{m_1 m_2} h_1^{\tilde{r}}$ and give $\tilde{r}$.
- c. Show that $h_1^{\tilde{r} q_1} = 1$.

Example — a multiplication in the toy instance

Model: $\mathbb{G}_1$ is the subgroup of order 35 in $\mathbb{Z}_{211}^*$, $g_1 = 5$, and $e(g^a, g^b) = g_1^{ab \bmod 35}$ (bilinear, but not efficient).

- a. Compute $h_1 = g_1^{\alpha q_2}$ and $g_1^{q_1}$ modulo 211.
- b. $C_1 = 50 = g^{22}$, $C_2 = 10 = g^{29}$. With $r = 11$, compute $C = e(C_1, C_2) h_1^r$.
- c. Decrypt C . Check against $\tilde{r}$ from the previous example.

Example — what BGN can compute

On encrypted bits $x_1, \dots, x_k$ and $y_1, \dots, y_k$, which can BGN evaluate?

- a. $x_1 + x_2 + \dots + x_k$
- b. $x_1 x_2 + x_3 x_4$
- c. $\sum_i x_i y_i$
- d. $x_1 x_2 x_3$
- e. For those it can, the condition on k and q_2 .

Subgroup decision assumption

Given $(n, \mathbb{G}, \mathbb{G}_1, e)$ but not q_1, q_2 , no efficient algorithm distinguishes a uniform element of $\mathbb{G}$ from a uniform element of the order- q_1 subgroup.

Under it BGN is IND-CPA secure (Lecture 8): h^r looks uniform in $\mathbb{G}$, so $g^m h^r$ hides m .

Factoring n breaks it, so BGN is at most as hard as factoring. DDH is not available — the pairing breaks it.

- ① A pairing $e(u^a, v^b) = e(u, v)^{ab}$ multiplies in the exponent once; it also makes DDH easy.
- ② BGN: $n = q_1 q_2$, $h = u^{q_2}$ of order q_1 ; $C = g^m h^r$, $\text{Pk} = (n, \mathbb{G}, \mathbb{G}_1, e, g, h)$, $\text{Sk} = q_1$.
- ③ $C^{q_1} = (g^{q_1})^m$ kills the mask exactly; a small discrete log recovers m , so $T < q_2$ and in practice $T = 1$.
- ④ $C_1 C_2$ adds messages, unlimited; $e(C_1, C_2) h_1^r$ multiplies them once, landing in $\mathbb{G}_1$. Degree-two polynomials only.
- ⑤ Security: subgroup decision, at most as hard as factoring n .